

Subrata Paul – Résumé



Bioinformatics Scientist

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Experience:

Nov 2021 Ongoing	Bioinformatics Scientist, Axle Informatics I work on multiple projects at the National Institute of Allergy and Infectious Diseases (NIAID), the National Institute of Arthritis and Musculoskeletal and Skin Diseases (NIAMS), and the National Heart, Lung, and Blood Institute (NHLBI). The projects include analyzing whole genome sequencing, bulk RNA-seq, single-cell RNA-seq, proteomics, survival analysis, machine learning, and data mining. Throughout these projects, I am contributing to the study of many diseases such as AIDS, rheumatoid arthritis, capillary syndrome, cryptococcal meningitis, T-cell exhaustion, and viral infections, including COVID-19.
Sep 2021 to Oct 2021	Bioinformatics Analyst III, Leidos Biomedical Research Inc. My primary responsibility is to provide statistical genetics and bioinformatics support to the projects at the NIAID and the NIAMS. The projects include the rare variant association of scleroderma and Systemic Capillary Leak Syndrome (SCLS) and the effect of arthritis on a patient's financial and social life.
Jan 2020 to Aug 2021	Postdoctoral Associate, Institute for Behavioral Genetics, CU Boulder In collaboration with multiple labs, I am working on uncovering genetic and phenotypic correlations among different pain conditions using UK Biobank data. In another project, I am investigating the performance of dosage data in estimating heritability compared to the hard-call genotype. I am co-supervising two graduate students in their projects on genetic genetic correlations of behavioral traits using a genetic relationship matrix-based approach and quantifying phenotypic assortment.
May 2015 to Dec 2019	Research Assistant, Dept. of Mathematical and Statistical Science, CU Denver I was involved in genetic risk prediction of generalized vitiligo. My responsibilities were data pre-processing, genome-wide association analysis, modeling polygenic risk scores, comparing multiple risk score models, dissect risk score models using functional annotation. I was also involved in analyzing secondary phenotypes and advanced method development to incorporate phenotypic heterogeneity into the association framework.
May 2018 to Jul 2018	Collaboration with Department of Electrical Engineering, CU Denver The Objective of this project was real-time electricity price forecasting to optimize electricity cost. I applied an artificial neural network on an intuitively designed feature space for forecasting electricity prices in real-time.
June 2017 to July 2017	Research Assistant, Dept. of Mathematical and Statistical Science, CU Denver A functional data analysis approach was developed to model spatiotemporal data. In this project, I used sparse matrices, numerical linear algebraic methods, and parallel computing to reduce computational expenses.

Publications and Posters

- 2026 Fernanda Gutierrez-Rodrigues, Emma M Groarke, Rui Miao, Bhavisha A Patel, **Subrata Paul**, et al. Clonal selection and evolution after treatment of severe aplastic anemia. *NEJM Evidence*, 5(5), 2026
- Amy P Hsu, **Subrata Paul**, et al. Gata2 at 14: genotype-phenotype correlations. *Haematologica*, 111(1), 2026
- 2025 Michael M Ward, Arthur Weinstein, Paloma Alejandro-Silva, Alan K Matsumoto, Rukmini Konatlapalli, Neil Stahl, Gail S Kerr, Jamal Mikdashi, Phillip Kempf, Nehal R Shah, **Subrata Paul**, et al. Discontinuation versus continuation of maintenance treatment with tumor necrosis factor inhibitors in patients with rheumatoid arthritis with low disease activity or remission: A randomized double-blind placebo-controlled trial. In *Seminars in Arthritis and Rheumatism*, page 152831. Elsevier, 2025
- Laura Lewandowski, Linda Hiraki, Christiaan Scott, Ana Barrera-Vargas, Jorge Romo Tena, Diana Gómez-Martín, Michael J Ombrello, Ivona Aksentijevich, Zuoming Deng, Anthony M Musolf, **Subrata Paul**, et al. Next generation sequencing analysis reveals complex genetic architecture of childhood-onset systemic lupus erythematosus. *Lupus Science & Medicine*, 12(1), 2025
- Manuel A Cintron, Yvonne Baumer, Alina PS Pang, Elizabeth M Aquino Peterson, Lola R Ortiz-Whittingham, Joshua A Jacobs, Sonal Sharda, Kameswari A Potharaju, Andrew S Baez, **Subrata Paul**, et al. Associations between the neural-hematopoietic-inflammatory axis and dna methylation of stress-related genes in human leukocytes: Data from the washington, dc cardiovascular health and needs assessment. *Brain, Behavior, & Immunity-Health*, page 100976, 2025
- Timothy Bhattacharyya, Gourinandan Saravanan, Suhyeon Yoon, and **Subrata Paul**. The missing revision burden: Total hip and knee replacement revision rates in the united states, 1996 to 2020. *JBJS Open Access*, 10(4):e25.00192, 2025
- Brice Calco, Colin L Sweeney, Joseph Graves Steiner, Tongguang Wang, Tovah E Markowitz, **Subrata Paul**, et al. A novel frameshift mutation in phosphoinositide 3-kinase regulatory subunit 1 (pik3r1) causes immunodeficiency and amyotrophic lateral sclerosis (als). *bioRxiv*, 2025
- Nicholas C Lee, Fernanda Gutierrez-Rodrigues, Ruba Shalhoub, Tania Machado, Jennifer Lotter, Lemlem Alemu, Diego Quinones Raffo, **Subrata Paul**, Sachiko Kajigaya, Geraldine Aubert, et al. Danazol treatment for telomere biology disorders: long-term results of a phase 1/2 study. *Blood Advances*, 10(1):208–212, 2025
- Andrea C Pichler, Jennifer L Cannons, Dominic P Golec, Julie M Reilley, Dan Corral, Eduard Ansaldo, Qin Xu, **Subrata Paul**, et al. Pi3k δ promotes t cell effector differentiation and plasticity during chronic infection. *bioRxiv*, 2025
- 2024 Laura Lewandowski, Linda Hiraki, Christiaan Scott, Ana Barrera-Vargas, Diana Gomez-Martin, Michael Ombrello, Ivona Aksentijevich, Zuoming Deng, Anthony Musolf, **Subrata Paul**, et al. Next generation sequencing analysis reveals complex genetic architecture of childhood-onset systemic lupus erythematosus. In *ARTHRITIS & RHEUMATOLOGY*, volume 76, pages 3638–3640. WILEY 111 RIVER ST, HOBOKEN 07030-5774, NJ USA, 2024
- Monishankar Sarkar, **Subrata Paul**, and Joydeb Garai. Climate-induced coastal occupational vulnerability and livelihood insecurity: Insights from coastal bangladesh. *Progress in Disaster Science*, 24:100382, 2024

- Fernanda Gutierrez-Rodrigues, Emma M Groarke, Natthakan Thongon, Juan Jose Rodriguez-Sevilla, Luiz Fernando B Catto, Marena R Niewisch, Ruba Shalhoub, Lisa J McReynolds, Diego V Clé, Bhavisha A Patel, **Subrata Paul**, et al. Clonal landscape and clinical outcomes of telomere biology disorders: somatic rescue and cancer mutations. *Blood*, 144(23):2402–2416, 2024
- Ararat J Ablooglu, Wei-Sheng Chen, Zhihui Xie, Abhishek Desai, **Subrata Paul**, Justin B Lack, Linda A Scott, A Robin Eisch, Arkadiusz Z Dudek, Samir M Parikh, et al. Intrinsic endothelial hyperresponsiveness to inflammatory mediators drives acute episodes in models of clarkson disease. *The Journal of Clinical Investigation*, 134(10), 2024
- 2023 Mariana Correia Marques, **Subrata Paul**, Carol Lake, Ly-Lan Bergeron, Rashmi Sinha, Luciana Peixoto, Marinka Twilt, and Michael J Ombrello. Online survey of covid-19 immunization and infection in patients with systemic juvenile idiopathic arthritis and adult-onset still's disease. *Pediatric Rheumatology*, 21(1):139, 2023
- Amit Patel, Jules D Allbritton-King, **Subrata Paul**, and Timothy Bhattacharyya. Bone health is improving over time: data from framingham cohorts. *Archives of Osteoporosis*, 18(1):119, 2023
- 2022 M. Ali Rai, Victoria Shi, Brooke D. Kennedy, Emily J. Whitehead, Jesse S. Justement, **Subrata Paul**, Jana Blazkova, and Tae-Wook Chun. Effect of aging on immune cells in male hiv-infected and -uninfected healthy individuals. *AIDS*, 2022
- Danielle Niemann, **Subrata Paul**, and Humairat H Rahman. Avalanche preparedness and accident analysis among backcountry skier, sidecountry, and snowmobile fatalities in the united states: 2009 to 2019. *Wilderness & Environmental Medicine*, 33(2):197–203, 2022
- Alex Pu, **Subrata Paul**, Brian Luke, and Timothy Bhattacharyya. Better mobility is associated with higher incomes and longer working years among older adults. *Clinical Orthopaedics and Related Research*, pages 10–1097, 2022
- 2020 **Subrata Paul** and Stephanie A Santorico. Optimized genetic risk prediction for vitiligo and its use to define disease subtypes. In *GENETIC EPIDEMIOLOGY*, volume 44-5, page 515, 2020
- Subrata Paul** and Stephanie A Santorico. Incorporation of heterogeneity in a case-control study through a mixture model. *bioRxiv*, 2020
- 2019 Genevieve H. L. Roberts, **Subrata Paul**, Daniel Yorgov, Stephanie A. Santorico, and Richard A. Spritz. Family clustering of autoimmune vitiligo results principally from polygenic inheritance of common risk alleles. *The American Journal of Human Genetics*, July 2019
- 2018 Md Habib Ullah, **Subrata Paul**, and Jae-Do Park. Real-time electricity price forecasting for energy management in grid-tied mtmc microgrids. In *2018 IEEE Energy Conversion Congress and Exposition (ECCE)*, pages 73–80. IEEE, 2018
- Subrata Paul** and Stephanie A Santorico. Incorporation of heterogeneity through a mixture model to boost power of association tests. In *GENETIC EPIDEMIOLOGY*, volume 42-7, pages 722–722, 2018
- 2016 Stephanie A Santorico, **Subrata Paul**, Daniel Yorgov, Ying Jin, Tracey Ferrara, and Richard A Spritz. A comparison of genetic risk prediction and subtyping for generalized vitiligo. In *GENETIC EPIDEMIOLOGY*, volume 40-7, pages 657–658, 2016
- 2015 Stephanie A Santorico, Ying Jin, Daniel Yorgov, **Subrata Paul**, Tracey Ferrara, and Richard Spritz. Optimized genetic risk prediction for vitiligo and its use to define disease subtypes. In *GENETIC EPIDEMIOLOGY*, volume 39-7, pages 577–578, 2015
- 2009 Sheik Ahmed Ullah, **Subrata Paul**, and Md. Sharif Ullah Mozumder. On information carriage through sigma-algebra in binomial asset pricing model. *Journal of Statistical Studies*, 28:1–8, 2009

Education

Aug 2014 to Dec 2019	University of Colorado Denver PhD in Applied Mathematics Concentration in Statistical Genetics. Through courses, I have learned advanced statistical models, Bayesian analysis, machine learning techniques, and human genetics etc.
Aug 2012 to May 2014	Ball State University, Indiana, USA MS in Mathematics MS thesis: <i>Numerical Multigrid algorithm for solving Integral Equation</i> . Focused on solving special types of integral equations with less computational complexity using multigrid method and to show its efficiency.
Aug 2008 to May 2010	University of Dhaka MS in Applied Mathematics MS thesis on numerical integration.
Mar 2003 to May 2008	University of Dhaka BS in Mathematics with minors in Computer Science, Statistics and Physics BS project: <i>On Binomial Asset Pricing Model with convergence to Black-Scholes model</i> . Discussed the power of binomial model on asset pricing and proven its strength by showing its convergence to Black-Scholes model.

Skills

Core	Statistical Modeling Human Genetics scRNA-seq ATAC-seq Proteomics Survival Analysis Machine Learning Computational Math	Programming Languages	R (Expert) Python (Advanced level) SQL, MATLAB, C++, FORTRAN (Intermediate)
Languages	English (Proficient) Bengali (Native) Hindi (Communication)	Others	Latex, Beamer Mathematica Data Visualization Inkscape

Referees

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